AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2016

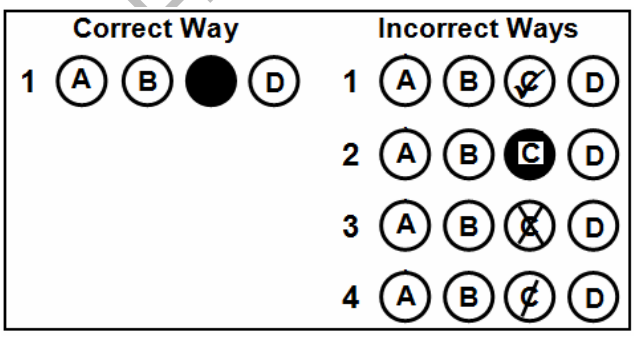
Physics Paper I

Time: 35 minutes Marks: 25

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. If the frequency of a wave is doubled, then the time period of the wave will

   A. become double.
   B. decrease to half.
   C. remain the same.
   D. decrease to a quarter.

2. If the velocity of a sound wave is 340 m/s and its frequency is 200 Hz, then the wavelength of this sound wave will be

   A. 0.6 m
   B. 1.7 m
   C. 140 m
   D. 540 m

3. When sound travels through air, the air particles vibrate

   A. in random direction.
   B. along the direction of wave propagation.
   C. perpendicular to the direction of wave propagation.
   D. parallel and perpendicular to the direction of wave propagation.

4. A sound wave is a mechanical wave. This means that

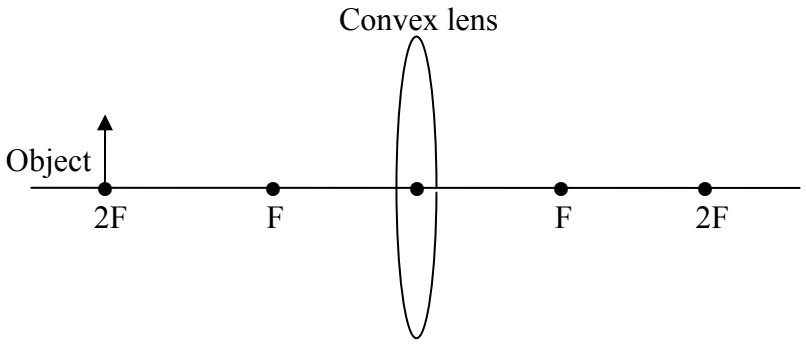
   A. sound wave transports energy through vacuum.
   B. particles of the medium move with the speed of light.
   C. a medium is required for sound wave to transport energy.
   D. particles of the medium move perpendicular to the direction of motion.

5. If an object is placed at 8 cm in front of a concave mirror and the focal length of the mirror is 4 cm, then the image will be formed at

   A. 2 cm
   B. 4 cm
   C. 8 cm
   D. 12 cm

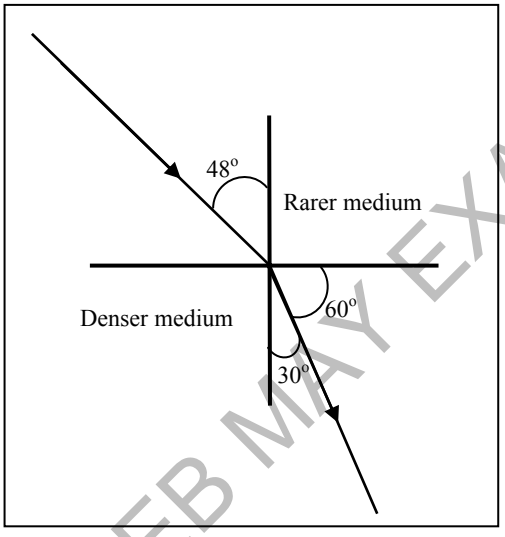
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6. Where will an image form in the given diagram if the magnification is 1?



A. At F
B. Between F and 2F
C. At 2F
D. Beyond 2F

7. In the given diagram, the refractive index of the denser medium with respect to the rarer medium can be calculated by



A. $\frac{\sin 48°}{\sin 30°}$
B. $\frac{\sin 30°}{\sin 48°}$
C. $\frac{\sin 48°}{\sin 60°}$
D. $\frac{\sin 60°}{\sin 48°}$

8. An eye and a camera are similar in function because image formed by both are

A. virtual, erect and larger in size.
B. real, inverted and larger in size.
C. virtual, erect and smaller in size.
D. real, inverted and smaller in size.

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9. The force per unit charge is known as

   A. electric flux.
   B. electric current.
   C. electric potential.
   D. electric field intensity.

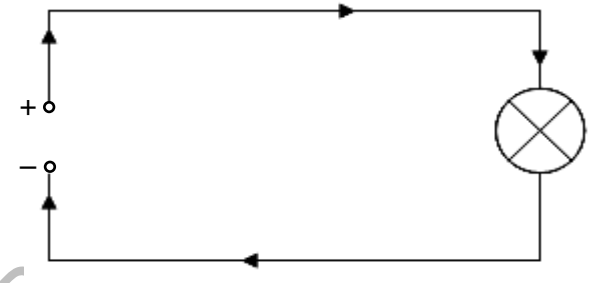
10. When three capacitors of different capacitance are connected in a parallel combination, then the total capacitance of the combination will be

   A. lesser than the individual capacitance.
   B. greater than the individual capacitance.
   C. the sum of the capacitance of any two capacitors.
   D. the product of the capacitance of all the three capacitors.

11. If the electric current through a resistor is 2 A and potential difference across the resistor is 6 V, then the value of power dissipated in the resistor will be

   A. 3 W
   B. 4 W
   C. 8 W
   D. 12 W

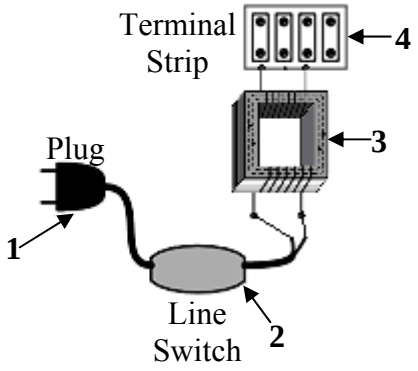
12. In the given circuit diagram, a 100 W bulb is operated with 220 V source, the current passing through the bulb is



   A. (100 × 220) A
   B. (220 / 100) A
   C. (100 / 220) A
   D. (220 + 100) A

13. Which of the following devices is shown in the given diagram at position **3**?

   A. A resistor
   B. A capacitor
   C. A generator
   D. A transformer



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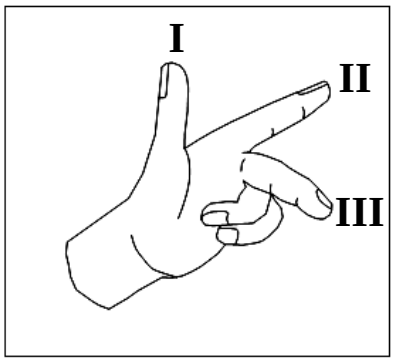
14. The working of an AC generator is an example of

A. inductor.
B. impedance.
C. mutual induction.
D. electromagnetic induction.

15. If a piece of copper wire is placed perpendicular to a magnetic field and the current in the wire increases, then the force applied by the magnetic field on the wire will

A. increase.
B. decrease.
C. become zero.
D. remain the same.

16. In the given figure, which of the following combinations correctly represents Fleming's left hand rule?



	I	II	III
A	Field	Force	Current
B	Current	Force	Field
C	Force	Field	Current
D	Force	Current	Field

17. In a logic gate, the total number of truth table entries for a two-inputs circuit are

A. 2
B. 4
C. 8
D. 16

18. Which of the following logic gates is used to invert the output?

A. OR gate
B. NOT gate
C. NOR gate
D. NAND gate

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19. When a radio wave is intercepted by an antenna. The antenna converts the wave into a/ an

    A. digital signal.
    B. analog signal.
    C. electrical signal.
    D. electromagnetic signal.

20. To collect information for a special purpose, store it in a computer and retrieve when needed is called

    A. data managing.
    B. word processing.
    C. communication technology.
    D. monitoring and controlling data.

21. The process of splitting U-235 by the bombardment of neutron into Ba-141 and Kr-92 is known as

    A. fusion reaction.
    B. fission reaction.
    C. physical change.
    D. chemical reaction.

22. If the half-life of an isotope is 1 day, then the quantity of that isotope in proportion to its initial quantity after 2 days will be

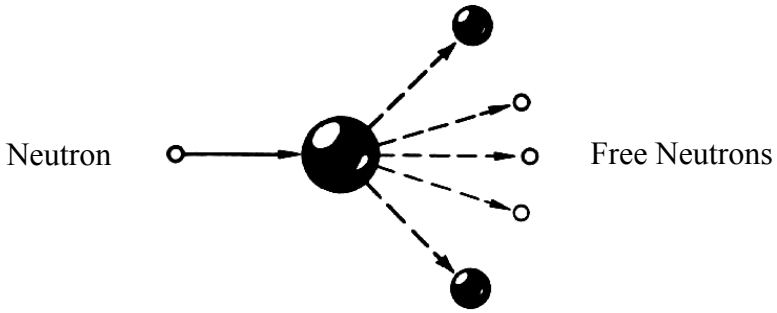
    A. 0
    B. $\frac{1}{2}$
    C. $\frac{1}{4}$
    D. $\frac{1}{8}$

23. The ionizing power of gamma rays is

    A. greater than beta particles.
    B. greater than alpha particles.
    C. less than both alpha and beta particles.
    D. less than alpha but greater than beta particles.

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24. Which of the following reactions is shown in the given diagram?



A. Nuclear fission
B. Nuclear fusion
C. Capturing reaction
D. Rearrangement reaction

25. Which of the following presents the harmful effect on human beings due to the prolonged exposure to nuclear radiations?

A. Heart attack
B. Throat infection
C. Genetic mutation
D. Brain haemorrhage

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END OF PAPER

Please use this page for rough work

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